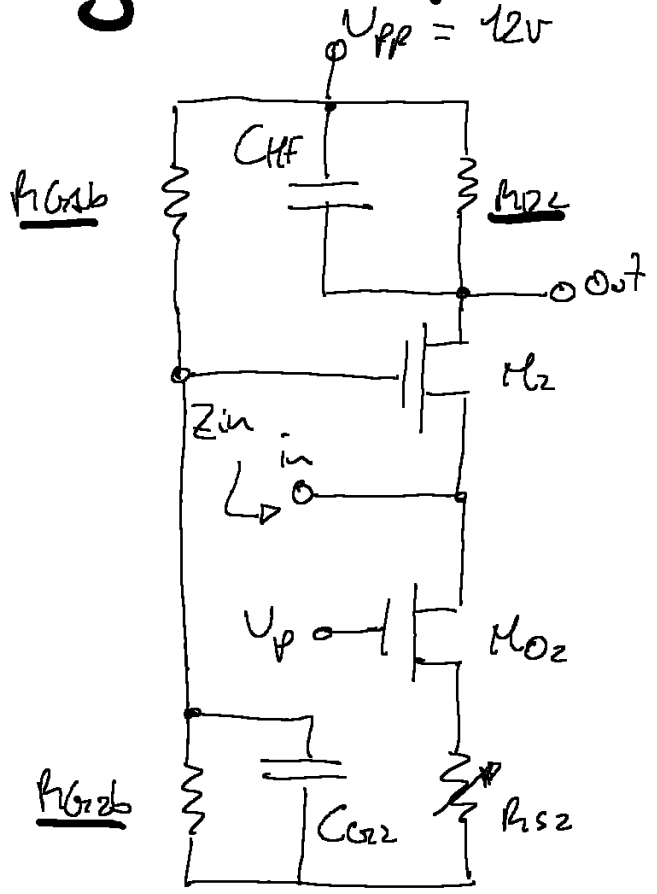
Segunda etapa



$R_{G1b} = 2M\Omega$

$R_{G2b} = 1M\Omega$

$R_{D2} = 1,5k\Omega$

$I_{DS} = k(V_{GS} - V_T)^2$

$V_T = 2V$

$r_{dc(on)} = 3,2\Omega$

a $V_{GS} = 5V$

$I_D = 0,05A$

$T < 125°$

$g_{m\,min} = 80$

a $V_{DS} = 10V$

$I_D = 0,2A$

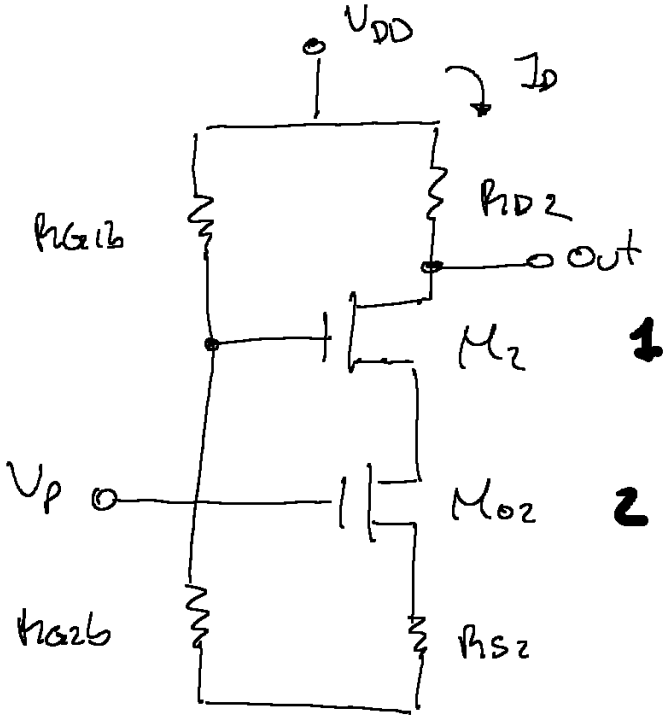
Cálculo de k

$I_{D(on)} = 1$ amper (típico) a $V_{DS} = 10V$ $V_{GS} = 10V$

$$k = \frac{I_{D(on)}}{(V_{GS} - V_T)^2} = \frac{1}{64} = 15,625 \frac{mA}{V^2}$$ **MO**

V_{SO} $V_{GS} = 4,5V$

$$k = \frac{I_{D(on)}}{(V_{GS} - V_T)^2} = 0,056 \frac{mA}{V^2}$$ com $V_{DS} = 10V$ $V_{GS} = 4,5V$



Quiero I_{DS}

Necesito calcular R_{S2}

Valores

V_{DD}

R_{G1b}

R_{G2b}

R_{D2}

K

V_T

Incógnitas

1 I_D ✓

2 V_{GS1} ✓

3 V_{GS2} ✓

4 V_{DS_2} ✓

5 V_{DS1} ✓

6 V_{D2} ✓

~~7 R_{S2}~~

8 V_{D1}

9 V_{S1}

10 V_{S2}

1,2 $I_{DS} = K(V_{GS1} - V_T)^2 = K(V_{GS2} - V_T)^2$

2 $V_{GS2} = V_p - I_D R_{S2}$

3 $V_{GS1} = V_{DD} \frac{R_{G2}}{R_{G1} + R_{G2}} - V_{D2}$

4 $V_{D2} = V_{DS_2} - I_D R_{S2}$

5 $V_{DD} - I_D(R_{D2} + R_S) - V_{DS1} - V_{DS2} = 0$

6 $V_{DS1} = V_{D1} - V_{S1}$

7 $V_{DS2} = V_{D2} - V_{S2}$

8 $V_{S2} = I_D R_S$

9 $V_{D1} = V_{DD} - I_D R_D$

$V_{S1} = V_{DS2} + I_D R_{S2}$

$$V_{DL} = V_{DD} - I_D R_{d2}$$

$$V_{S1} = V_{DS2} + I_D R_{S2}$$

$$V_{DS1} = V_{DD} - V_{DS2} - I_D(R_{d2} + R_{S2})$$

$$V_{DS1_max} = V_{DD} - V_{DS@2}$$

$$V_{DS1_opt} = \frac{V_{DD} - V_{DS@2}}{2}$$

¿Está bien esta condición?

$$V_{DS1_opt} = \frac{V_{DD} - V_{DS@2} - I_{D@}(R_{d2} + R_{S2})}{2}$$

¿O es así?

Si: $V_{DS1} + V_{DS2} = 12V$ → corte

Si: $V_{DS1} + V_{DS2} = 0$ → Sat

$V_{DS1} = V_{D1} - V_{S1}$ $\quad V_{S1} = V_{D2}$

$V_{DS2} = V_{D2} - V_{S2}$ $\quad V_{S2} = I_D R_{S2}$

$V_{D2} = V_{DS2} + V_{S2} = V_{S1}$ $\quad -(V_{D1} - V_{DS2} + V_{S2}) - V_{DS2}$

$V_{DS1} = V_{D1} - V_{DS2} + V_{S2}$ $\quad -V_{D1} + \cancel{V_{DS2}} - V_{S2} - \cancel{V_{DS2}}$

$-V_{D1} - V_{S2}$

$-V_{D1} - I_D R_{S2}$

$$g_m = \frac{\partial I_{DS}}{\partial V_{GS}}$$

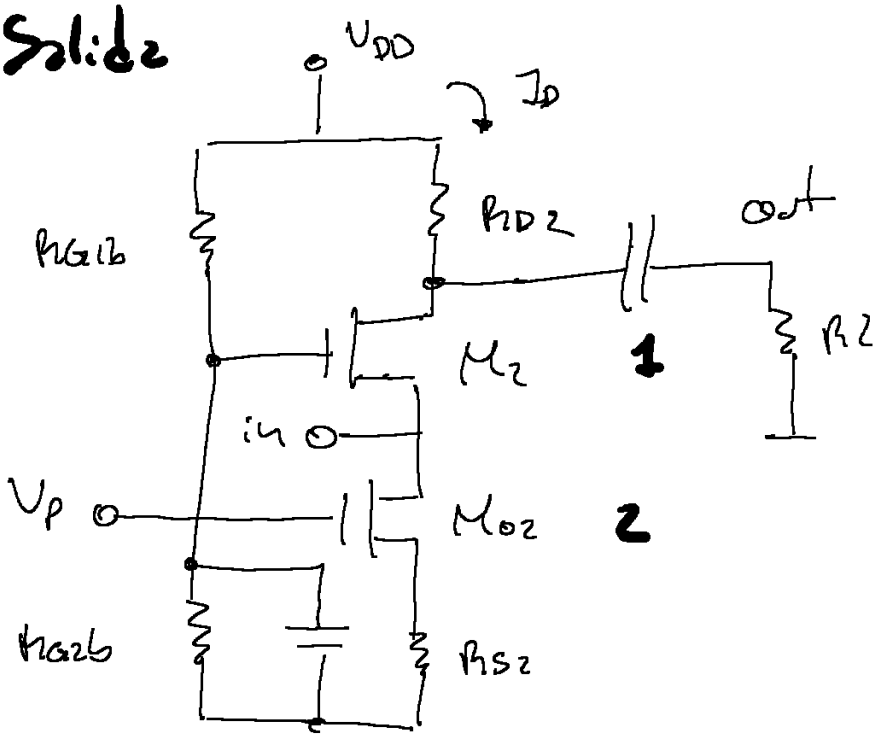
$$g_{m1} = 2k(V_{GS} - V_T)\Big|_{V_{GS} =}$$

=

$$g_{m2} = 2k(V_{GS} - V_T)\Big|_{V_{GS} =}$$

Rectas de Carga

Salida



$R_{CC} = R_{D2} + R_{S2}$

$R_{C2} = R_{D2} // R_L + R_{S2}$

$R_L = 1200$

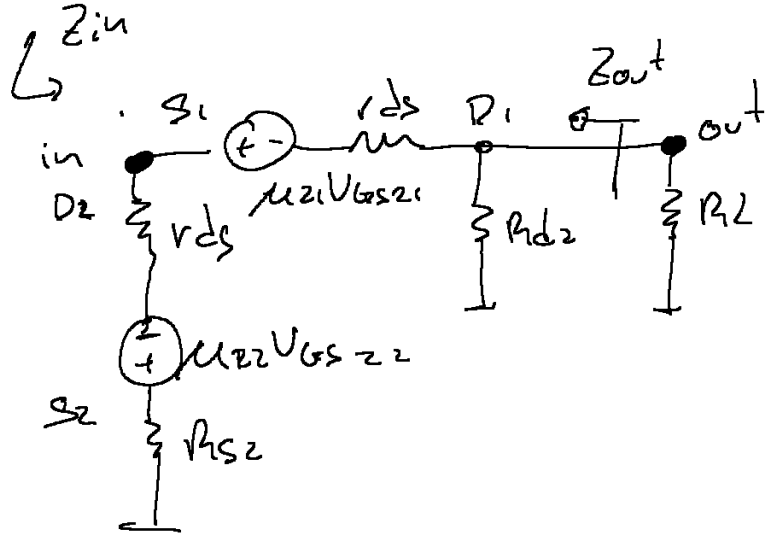
$$V_{DD} - V_{DS1} - V_{DS2} - I_{DS}(R_{d2} + R_{S2}) = 0$$

$$I_{DS_1} = \frac{V_{DD} - V_{DS1} - V_{DS2Q}}{R_{d2} + R_{S2}} \qquad P_{2R} \quad M_1$$

$$i_{DS_1} = I_{DS_1Q} - \frac{V_{DS} - V_{CC} - V_{DS2Q}}{R_{CA}}$$

$$I_{DS_2} = \frac{V_{DD} - V_{DS2} - V_{DS1Q}}{R_{CC}} \qquad P_{2R} \quad M_2$$

$$i_{DS_2} = I_{DS_2Q} - \frac{V_{DS} - (V_{CC} - V_{DS1Q})}{R_{CE}}$$



$g_m = 6{,}097\ mS$

$r_{ds} = 7{,}5\ M\Omega$

↳ tomo el máximo

$\mu = 0{,}045727S$

$Z_{out} = R_{d2} // \left[r_{ds} + r_{ds}(\mu_{21}+1) + R_{s2}(\mu_{22}+1)(\mu_{21}+1) \right]$

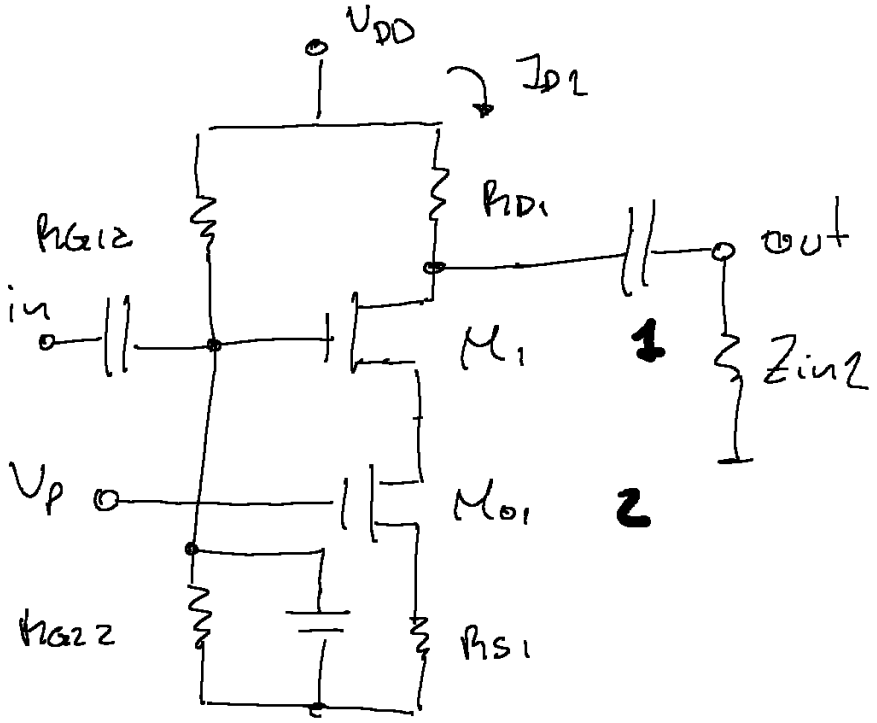
$= \cancel{1273{,}54\ M\Omega}$ ↓ $1162\ M\Omega \Rightarrow$ **RL = 1.2M**

Tomo $R_L = 1200\ M\Omega$ per MTE

$Z_{in} = \left(r_{ds} + \frac{R_{s2}}{1+\mu_{22}} \right) // \left(r_{ds} + R_{d2} // R_L \right)(\mu_{21}+1)$

$= \cancel{1120\ M\Omega}$ $556\ M\Omega$

Ejercicio 2



$$R_{cc} = R_{D1} + R_{S1}$$

$$R_{ca} = R_{D1} // Z_{in2} + R_{S1}$$

$$V_{DD} - V_{DS1} - V_{DS2} - I_{DS}(R_{d2} + R_{S2}) = 0$$

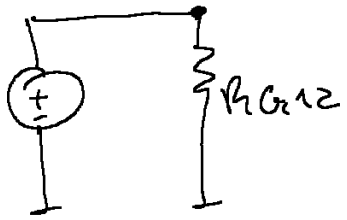
$$I_{DS_1} = \frac{V_{DD} - V_{DS1} - V_{DS2Q}}{R_{cc}} \qquad P_{22} \quad M_1$$

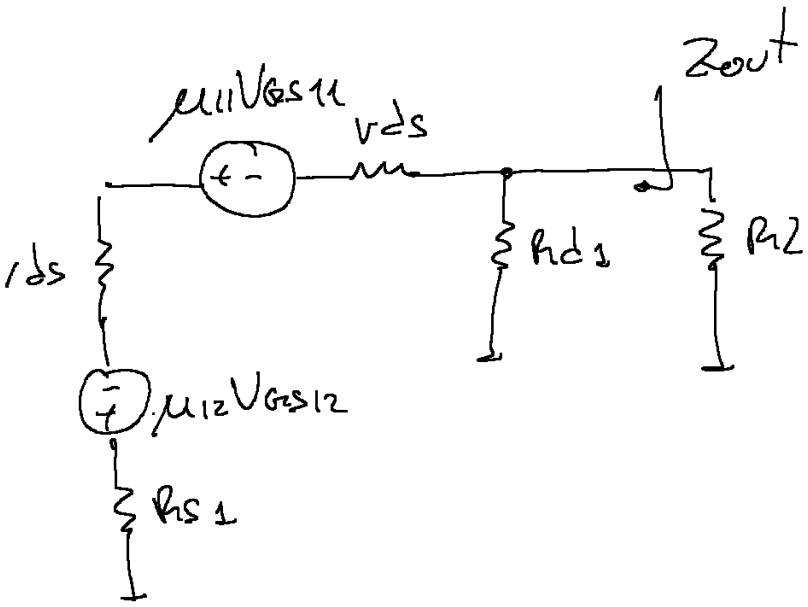
$$i_{DS_1} = I_{DS_1Q} - \frac{V_{DS} - V_{CC} - V_{DS2Q}}{R_{ca}}$$

$$I_{DS_2} = \frac{V_{DD} - V_{DS2} - V_{DS1Q}}{R_{cc}} \qquad P_{22} \quad M_2$$

$$i_{DS_2} = I_{DS_2Q} - \frac{V_{DS} - (V_{CC} - V_{DS1Q})}{R_{ce}}$$

Zin ↳ V_{GS1} $Z_{in} = R_{G12}$

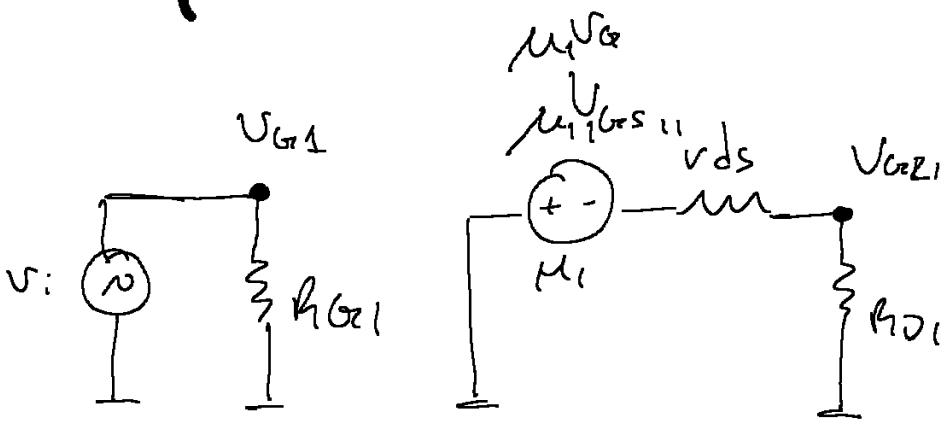




$$Z_{out} = R_{d1} // r_{ds} + [r_{ds} + R_{S1}(\mu_{12}+1)](\mu_{11}+1)$$

$$= 1738,5$$

Pequeña señal sin Cap



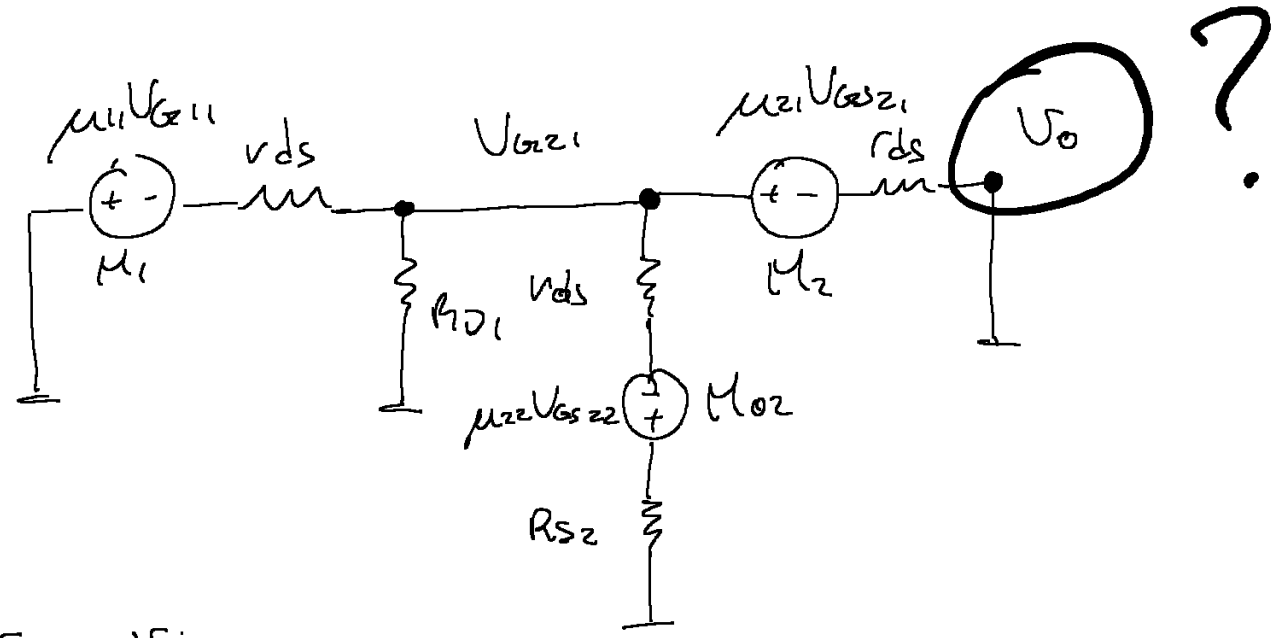
M_{01} no aparece

. V_G M_2 = 0 ?

ó

. V_G M_{02} = 0 ?

ó



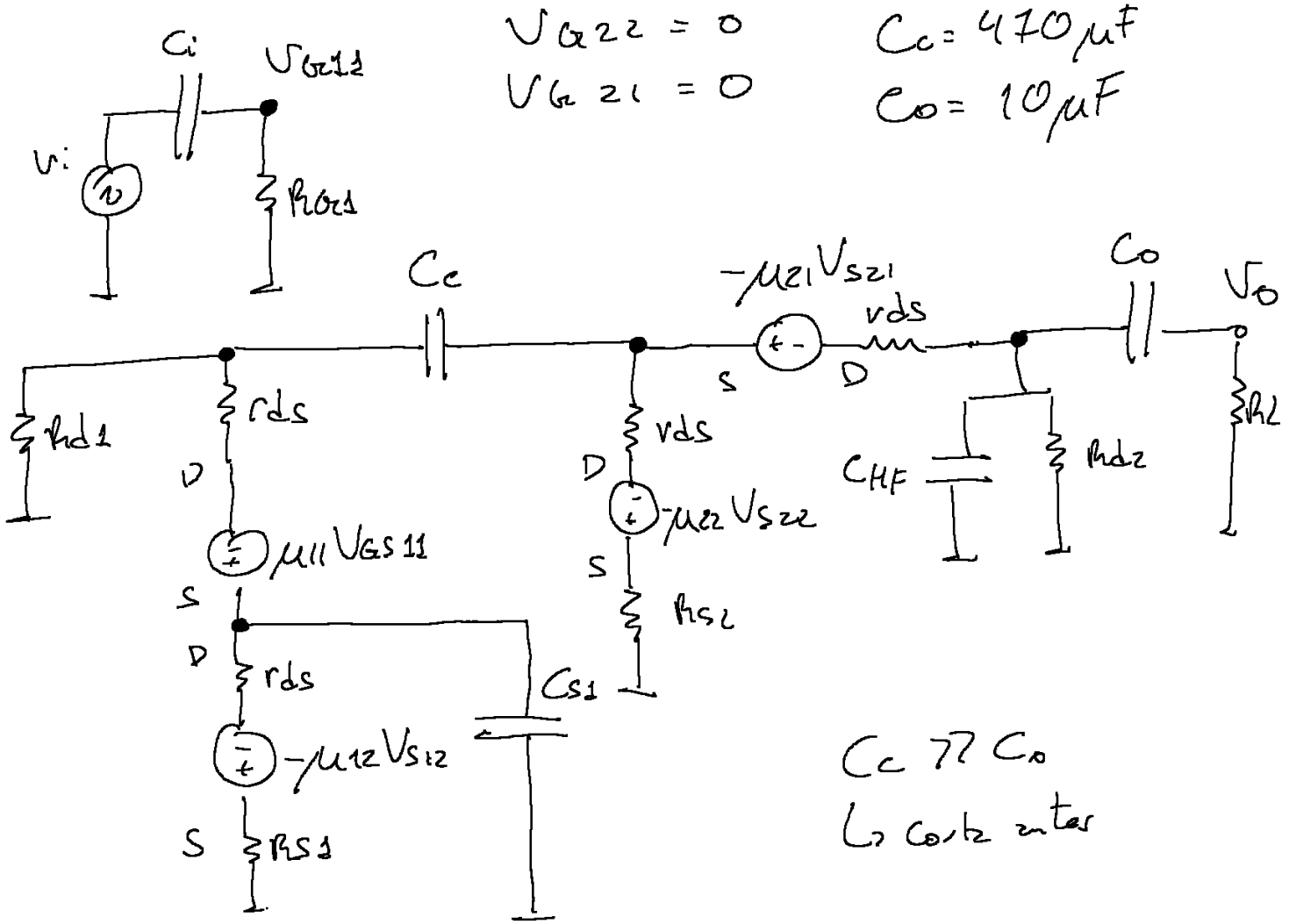
$V_{G11} = V_i$

$$\frac{V_{GZ1} + \mu_{11} V_{G11}}{r_{ds}} + \frac{V_{GZ1}}{R_{D1}} + \frac{V_{GZ1} + \mu_{22} V_{GS22}}{r_{ds} + R_{S2}} + \frac{V_{21} - \mu_{21} V_{GS21}}{r_{ds}} = 0$$

Pequeña Señal Con Cap

$V_{GG12} = 0$
$V_{GG22} = 0$
$V_{GG21} = 0$

$C_i = 10\mu F$
$C_c = 470\mu F$
$C_o = 10\mu F$



Cc ?? Co
↳ Corta antes

Para bajas frecuencias C_{HF} → abierto

Conozco Cc y Co

Cc y Co dan un cero en 0

Cs1 tiene un cero en ~~0~~

↳ 0 tiene Rth?

Ci → cero en 0

$R_{S1} = 6{,}8k\Omega$

$R_{S2} = 7{,}7k\Omega$

Polo en $\frac{1}{C_i R_{GG1}} = 0{,}15$ rad/s

Baja Frec

$\omega_{cs1} \gg \omega_{co}$ y ω_{cc}

Quiero que C_{s1} sea el dominante

Está abierto para C_c y C_o

$\therefore R_2? = 1.2k$

R_{TH} para C_{s1}

$$\left[r_{ds} + R_{s2}(\mu_{s+1})\right] // \left\{ r_{ds} + \left(r_{ds} + R_{s2}(\mu_{s2+1}) // \frac{\left[r_{ds} + (R_D // R_L)\right] // R_{d1}}{\mu_{2+1}} \right\} = 392,46\ \Omega$$

Como C_{s1} va a ser el dominante, está cortado para C_c y C_o

C_{HF} está siempre cortado

R_{TH} para C_c: asumo C_o cortado

1. $R_{d1} // r_{ds} + \left[r_{ds} + R_{s2}(\mu_{2+1})\right] // \frac{\left[r_{ds} + (R_2 // R_{d2})\right]}{\mu_{2+1}} = 52,01\ \Omega$

R_{TH} para C_o: asumo C_c abierto

2. $R_{d1} // r_{ds} + \left[r_{ds} + R_{s2}(\mu_{2+1})\right] // (r_{ds} + R_{d1}) = 1122,4\ \Omega$

R_{TH} para C_o: asumo C_c abierto

1. $R_L + R_{d2} // \left\{ r_{ds} + r_{ds}(\mu_{2+1}) + R_{s2}(\mu_{2+1}) \right\} = 23\Omega\ \Omega$

2. $R_L + R_{d2} // \left\{ r_{ds} + (\mu_{2+1}) \cdot \left[r_{ds} + R_{s2}(\mu_{2+1}) \right\} // r_{ds} // R_{d1} \right\}$ asumo C_o cortado

$= 1214,8\ \Omega$

Frecuencias de corte

$\mu_2 = 0,045912$ $\quad R_{S2} = 4,7k$

$\mu_1 = 0,036464$ $\quad R_{S1} = 7,7k$

De C_c

$$\omega_{C_c} = \frac{1}{C_c R_{TH1}} = 4,04 \text{ rad/s}$$

$$\omega_{C_c} = \frac{1}{C_c R_{TH2}} = 1,9 \text{ rad/s}$$

C_c abierto

De C_o

$$\omega_{C_o} = \frac{1}{C_o R_{TH1}} = 42,33 \text{ rad/s}$$

$$\omega_{C_o} = \frac{1}{C_o R_{TH2}} = 82,31 \text{ rad/s}$$

C_c cortado

Siempre están separados mínimo 1 década

Primero corta C_c en $\omega_{C_c} = 1,9 \frac{rad}{s}$ y

luego C_o en $82,31 \frac{rad}{s}$

$\omega_{C_{S1}} \leq 0,2 \text{ rad/s}$ para no ser el dominante

o $\omega_{C_{S1}} \geq 800 \text{ rad/s}$ para ser el dominante

$$\omega_{cs1} = \frac{1}{C_{s1} R_{TH}}$$

$$C_{s1} = \frac{1}{800 \frac{rad}{s} \cdot R_{TH}} = 3,2 \mu F$$

Cc con Co abierto

$$R_{TH} = R_{d1} // \left[r_{ds1} + r_{ds}^{0}(\mu_1+1) + R_{S1}(\mu_1+1)^2 \right] + \left[r_{ds_{02}} + R_{S2}\frac{(\mu_2+1)}{\mu_{02}+1} \right] // \frac{(r_{ds2}^2 + R_{d2})(\mu_2+1)^2}{(\mu_2+1)(\mu_{02}+1)}$$

Con Co cortado

$$R_{TH} = R_{d1} // \left[r_{ds1} + r_{ds}^{0}(\mu_1+1) + R_{S1}(\mu_1+1)^2 + \right] \left[r_{ds_{02}} + R_{S2}\frac{(\mu_2+1)}{(\mu_{02}+1)} \right] // \left[r_{ds_2} + (R_{d2} // R_2) \right] (\mu_2+1)^2$$

Co con Cc cortado

$$R_{TH} = R_2 + R_{d2} // \left\{ r_{ds2}^{2} + (\mu_2^{2}+1) \cdot \left[(r_{ds_{02}} + R_{S2}\frac{(\mu_2+1)}{\mu_{02}+1}) // R_{d1} // \left[r_{ds_1} + (\mu_1+1) \cdot \left[r_{ds_{01}} + R_{S1}\frac{(\mu_1+1)}{\mu_{01}} \right] \right] \right] \right\}$$

Con Cc abierto

$$R_L + R_{d2} // \left\{ r_{ds_2} + r_{ds_{02}}(\mu_2+1) + R_{S2}\frac{(\mu_2+1)^2}{\mu_2+1} \right\}$$

P2v2 Cs2 Cc, Co coletores

$$R_{TH} = \left[r_{ds1} + \left\{ R_{d1} \,//\, \left[\frac{r_{ds2} + R_{s2}(\mu+1)}{\mu_{o2}+1} \right] \right\} \,//\, \left[\frac{r_{ds2}\,(R_{d2} \,//\, R_2)}{\mu_{2}+1} \right] \right] \,//$$

$$\left(r_{ds\,o1} + R_{s1}(\mu_{o}+1) \right)^{\mu_1+1}$$

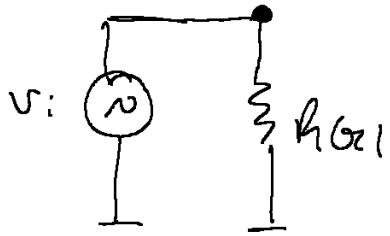
Ganancia en frecuencia media

$V_{GsMO2} = V_{GsMO2} = V_{GsM2} = 0$ $\quad R_L = 1{,}2k\Omega$

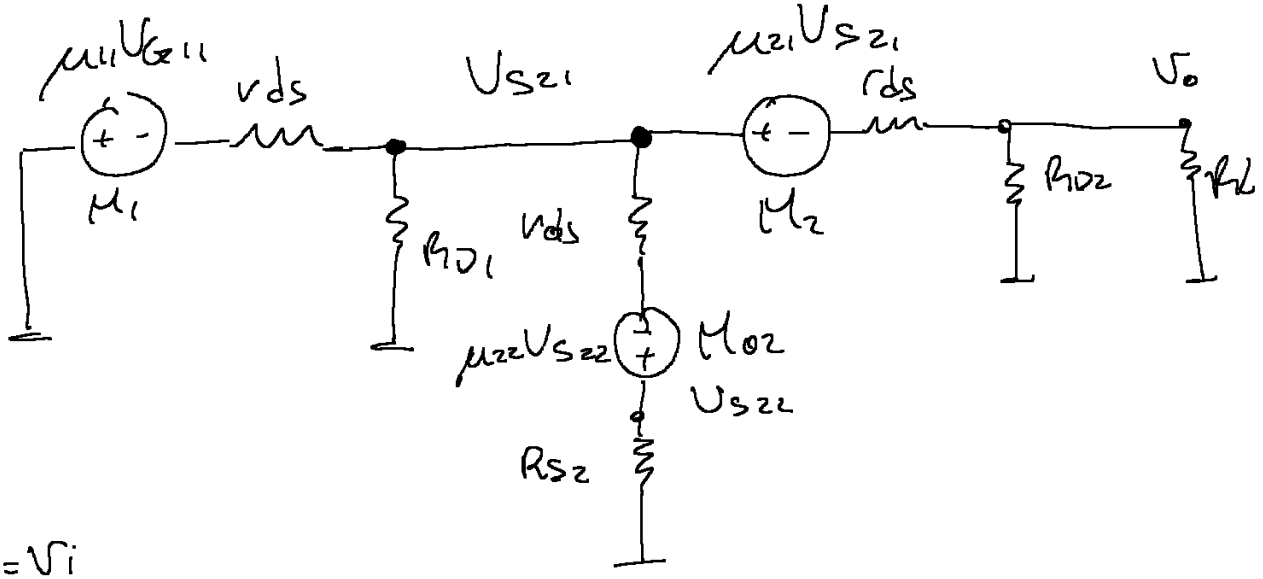
$\mu_2 = 0{,}045912$ $\quad R_{S2} = 4{,}7k$

$\mu_1 = 0{,}036464$ $\quad R_{S1} = 7{,}7k$

V_{G1} $\quad V_{SM_1} = 0$



$R_{G1} = R_{G12} // R_{G22}$



$V_{G1} = V_i$

$$\frac{V_{S21} + \mu_{11} V_{G11}}{r_{ds}} + \frac{V_{S21}}{R_{D1}} + \frac{V_{S21} + \mu_{22} V_{S22}}{r_{ds} + R_{S2}} + \frac{V_{S21} - \mu_{21} V_{S21} - V_o}{r_{ds}} = 0$$

$$V_o \left(\frac{1}{R_L} + \frac{1}{R_{D2}} \right) + \frac{V_o + \mu_{21} V_{GS21} - V_{GS21}}{r_{ds}} = 0$$

$$C_{S22} = \frac{V_{S21} + \mu_{22} V_{S22}}{r_{ds} + R_{S2}} \qquad V_{S22} = \frac{V_{S21} + \mu_{22} V_{S22}}{r_{ds} + R_{S2}} \cdot R_{S2}$$

$$V_{S22} = \frac{V_{S21} + \mu_{22} V_{S22}}{r_{ds} + R_{S2}} \cdot R_{S2}$$

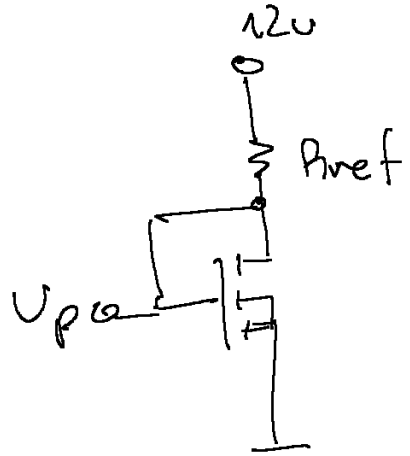
$$V_{S22} - \frac{\mu_{22} V_{S22} R_{S2}}{r_{ds} + R_{S2}} = V_{S1} \frac{R_{S2}}{r_{ds} + R_{S2}}$$

$$V_{S22} = V_{S1} \frac{R_{S2}}{r_{ds} + R_{S2}} \cdot \frac{1}{1 - \frac{\mu_{22} R_{S2}}{r_{ds} + R_{S2}}}$$

$$= \frac{V_{S1} R_{S2}}{r_{ds} + R_{S2} - \mu_{22} R_{S2}} = \frac{V_{S1} R_{S2}}{r_{ds} + R_{S2}(1 - \mu_{22})}$$

- ¿Que onda con los parámetros del transistor?
- ¿El polo de C_{S1} está en 0?
- ¿Está bien lo de la RL? Si
- ¿Que pasa con C_{GS2}?
- ¿Es correcto asumir V_{GSM01}; V_{GSM02}; V_{GSM2} y $V_{SM1} = 0$?
- ¿Para que está M real?
  ¿La Espejo de corriente
- ¿Cómo funcionan los espejos de corriente?

Espejo de Corriente



$$V_{cc} - R_{ref} I_{DS} - V_{DS} = 0$$

$$V_p = V_{GS} = V_G - V_S = V_G$$

$$I_{DS} = k (V_p - V_t)^2$$

$$V_{DS} = V_D - V_S = V_D$$

$$V_D = V_G = V_p$$

$$V_{GS} = V_G - V_S = V_G$$

$$V_{DD} - I_{DS} R_{ref} - V_p = 0$$

$$I_{DS} = k (V_p - V_t)^2$$

$$I_{DS} = \frac{V_{DD} - V_p}{R_{ref}}$$

1 Calcular la polarización con el espejo de corriente

Rs de cada M tiene que estar en el orden de los Ohm ✓ ✓

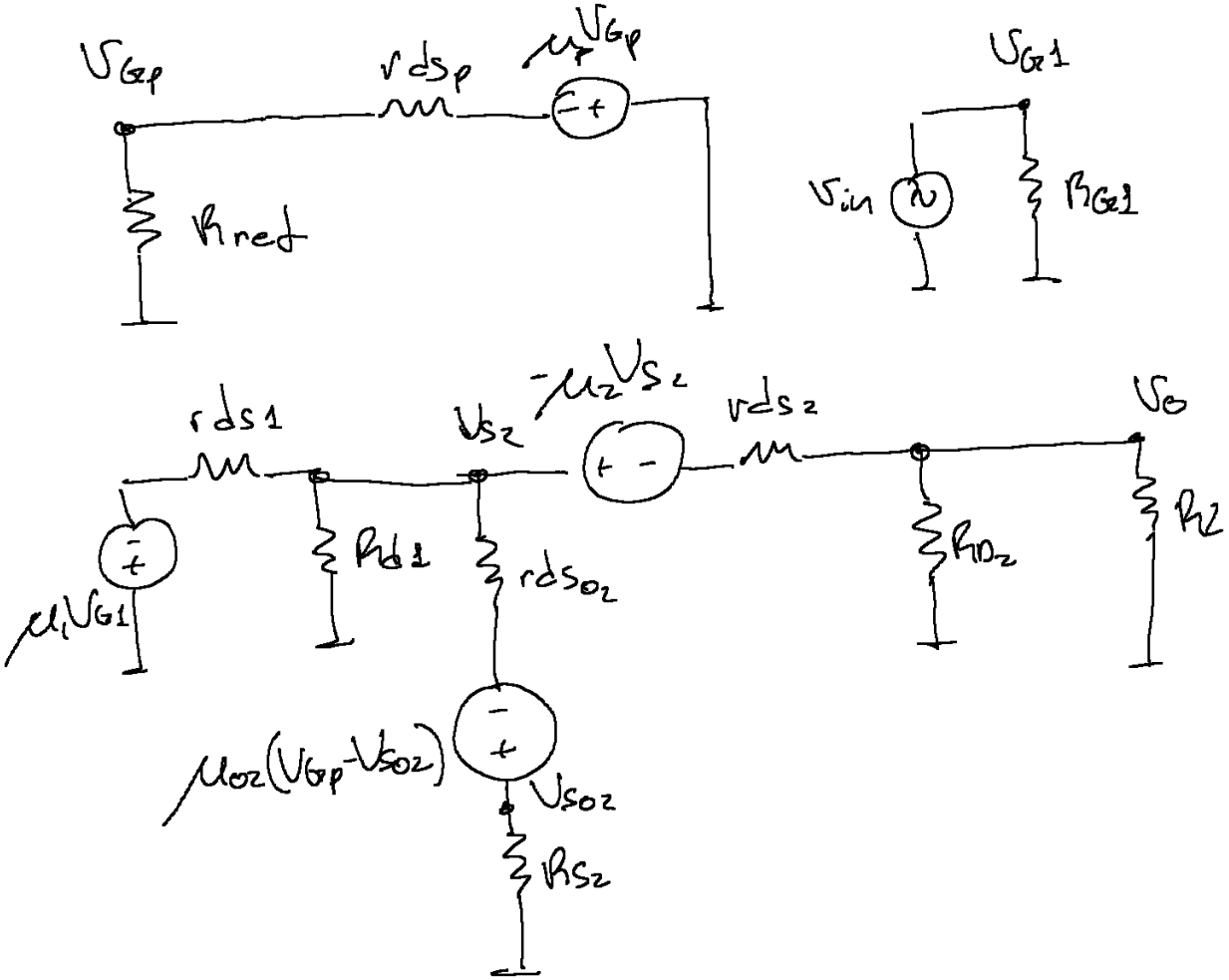
2 Calcular de nuevo la ganancia

3 Definir todos los polos

4 Calcular los límites de operación de cada etapa y del ampli

5 Hacer el Bode del ampli

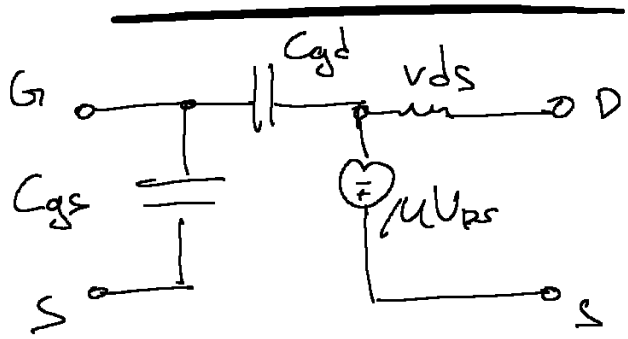
6 Avanzar informe


V_{Gp}
r_{dsp}
$\mu_p V_{Gp}$
R_{red}
V_{G1}
V_{in}
R_{G1}
r_{ds1}
V_{S2}
$-\mu_2 V_{S2}$
r_{ds2}
V_o
R_{d1}
r_{dso2}
R_{D2}
R_2
$\mu_1 V_{G1}$
$\mu_{o2}(V_{Gp}-V_{So2})$
V_{So2}
R_{S2}

Función de transferencia

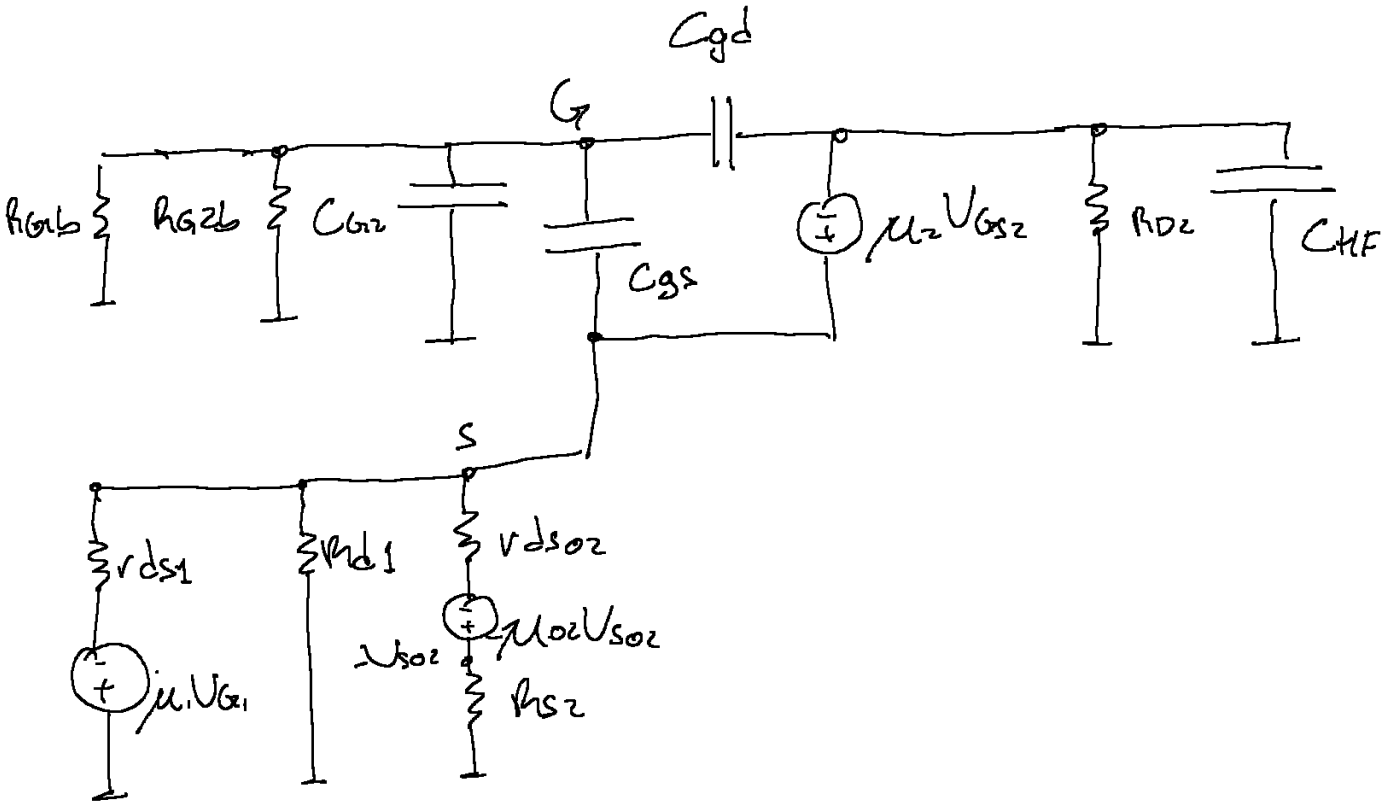
$$\mathcal{F}[T] = s^3 \left(1 + \frac{s}{\quad}\right.$$

Alta frecuencia

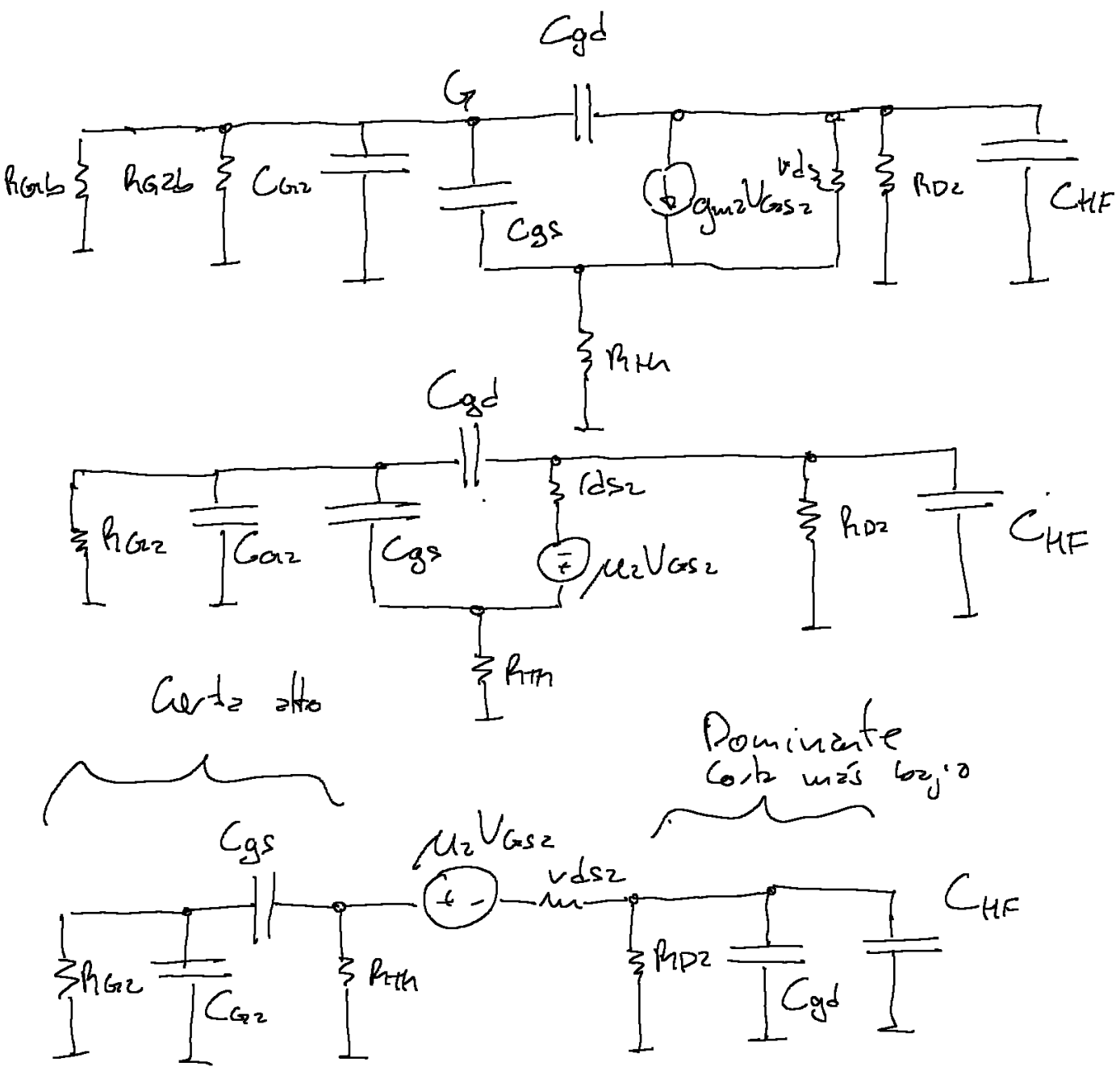


$C_{gd} =$

$C_{gs} =$

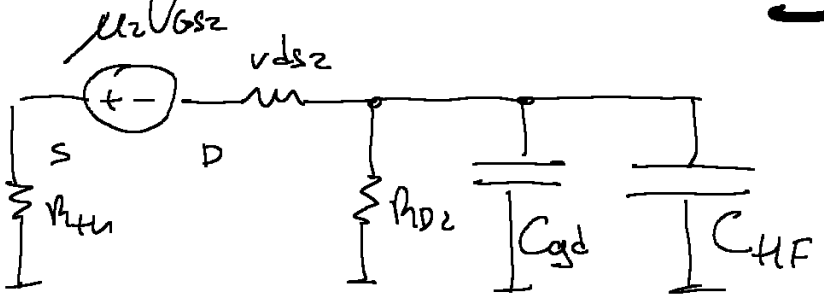


$$R_{TH} = r_{ds1} // R_{d1} // \left[r_{ds02} + (\mu_{02} + 1) R_{S2} \right]$$


Cgd
G
RG1b
RG2b
CG2
Cgs
gm2VGS2
vds
RD2
CHF
RTH
Cgd
RG2
CG2
Cgs
rds2
μ2VGS2
RD2
CHF
RTH
Cort2 alto
Dominante
Cort2 más bajo
Cgs
μ2VGS2
vds2
CHF
RG2
CG2
RTH
RD2
Cgd

Dominante

No Sirve



$$C_{eq} = C_{gd} + C_{HF}$$

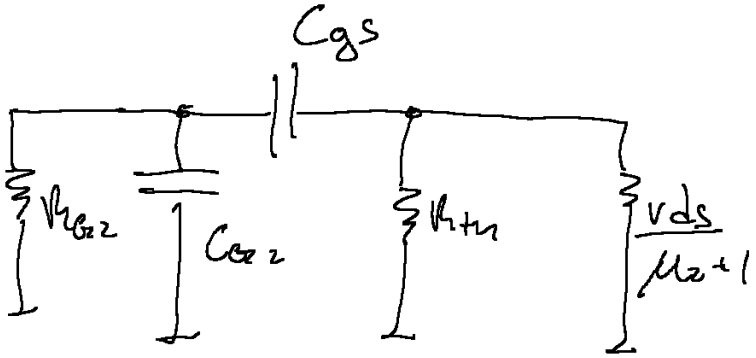
$$R_{eq} = R_{D2} // \left[v_{ds} + (\mu_2 + 1) R_{TH} \right]$$

$$\omega_{Zeq} = \frac{1}{C_{eq} R_{D2}}$$

$$\omega_{peq} = 20\,kHz$$

$$C_{eq} = \frac{1}{\omega_{peq} R_{eq}} = C_{gd} + C_{HF} \Rightarrow C_{HF} = C_{eq} - C_{gd}$$

No dominante



$$\omega_{Z_{G2}} = \frac{1}{C_{g2} R_{G2}}$$

$$\omega_{Z_{gs}} = 0$$

C_{G2} abierto

$$R_{gs} = R_{G2} + R_{TH} // \frac{v_{ds2}}{\mu_2 + 1}$$

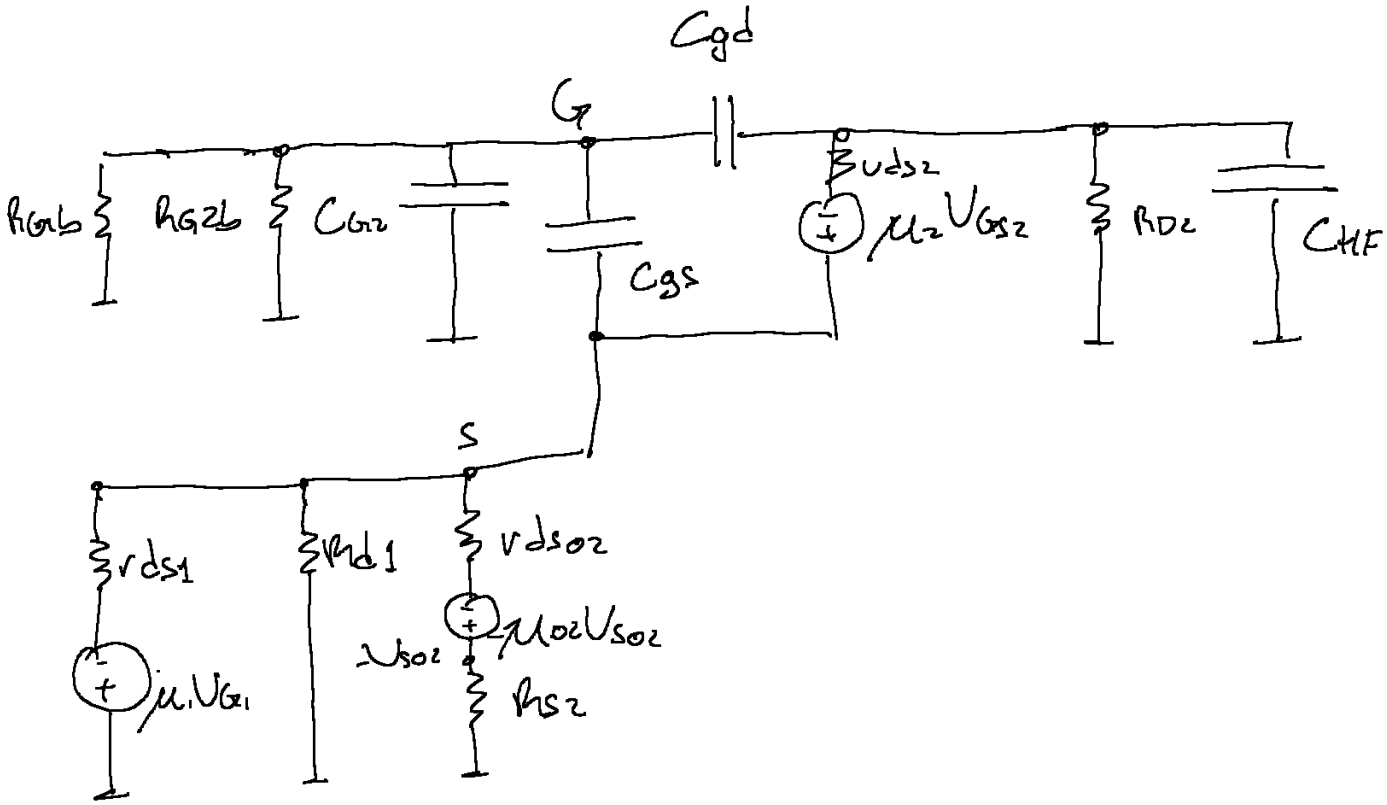
$$\omega_{pgs_1} = \frac{1}{C_{gs} R_{gs}} =$$

C_{G2} cerrado

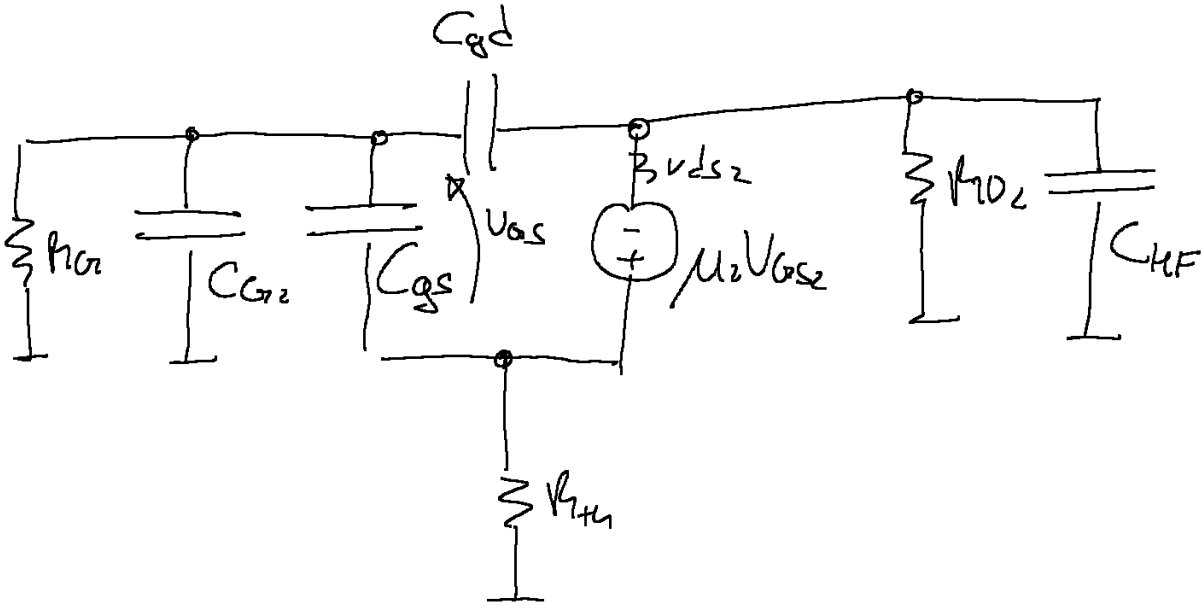
$$R_{gs} = R_{TH} // \frac{v_{ds2}}{\mu_2 + 1}$$

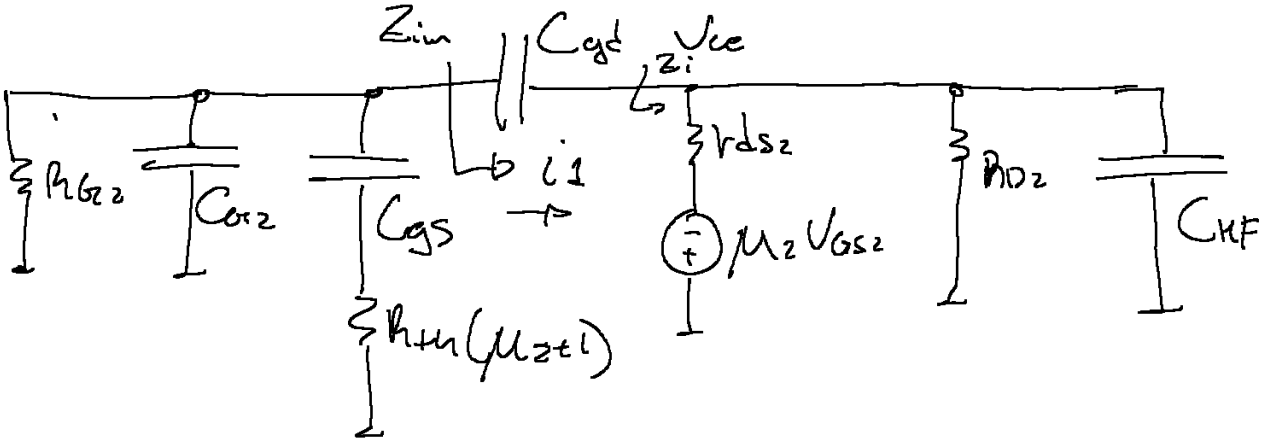
$$\omega_{pgs_2} = \frac{1}{C_{gs} R_{gs}} =$$

$$R_{G2} = R_{G2} // R_{TH} // \frac{v_{ds2}}{\mu_2 + 1}$$



$$R_{TH} = r_{ds1} // R_{d1} // \left[r_{ds02} + (\mu_{02} + 1) R_{S2} \right]$$





$$Z_i = \frac{1}{\frac{1}{r_{ds2}} + \frac{1}{R_{D2}} + sC_{HF}}$$

$$i_1 = (V_{gs2} - V_{ds2})\, sC_{gd}$$

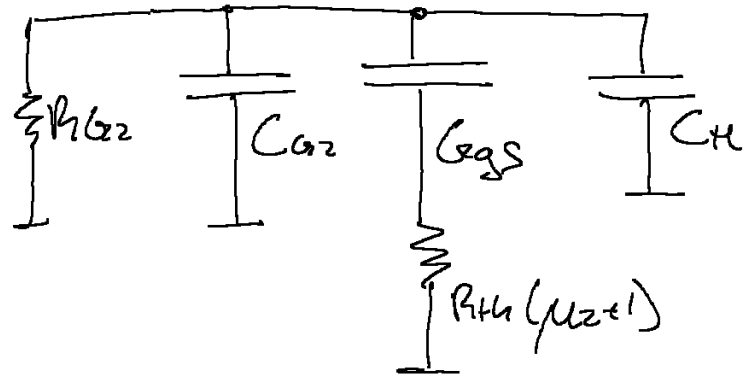
$$V_{ce} = -g_m V_{gs2} Z_i$$

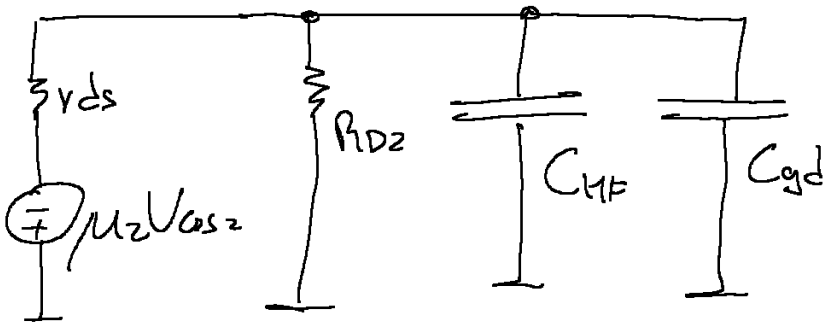
$$Y_{im} = \frac{1}{Z_{im}} = \frac{i_1}{V_{gs2}} = \frac{(V_{gs2} + g_m Z_i V_{gs2})\, sC_{gd}}{V_{gs2}}$$

$$Y_{im} = (1 + g_m Z_i)\, sC_{gd} \qquad R_{eq} = r_{ds} // R_{D2}$$

$$Z_i = \frac{R_{eq}\, sC_{HF}}{R_{eq} + sC_{HF}}$$

$$C_M = C_{gd}(1 + g_m Z_i)$$





$C_{eq} = C_{MF} + C_{gd}$

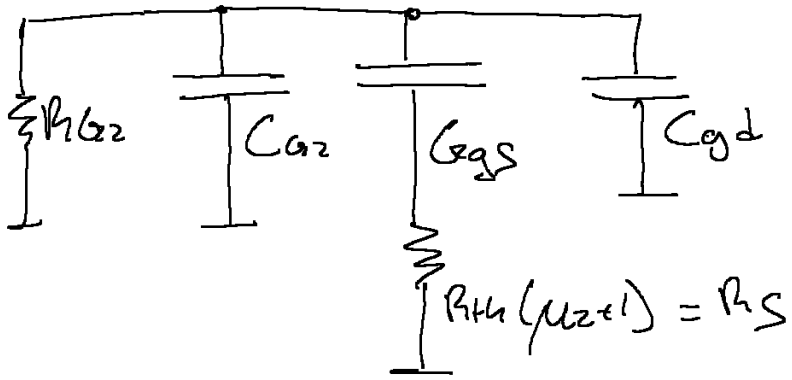
$R_{eq} = r_{ds2} // R_{D2}$

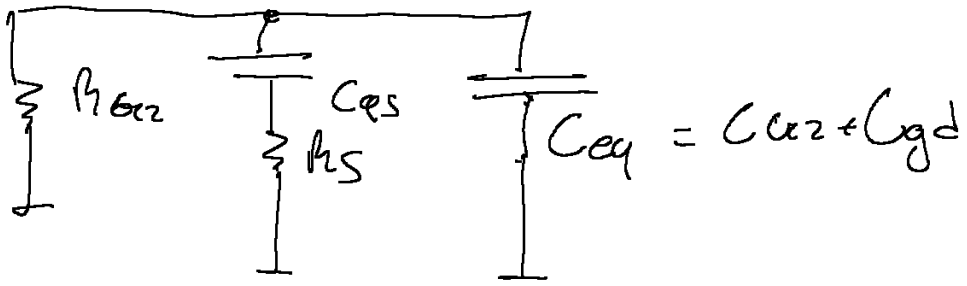
$\omega_{Zeq} = \frac{1}{C_{eq} R_{D2}}$

$\omega_{peq} = 20MHz$

$C_{eq} = \frac{1}{\omega_{peq} R_{eq}} =$

$C_{MF} = C_{eq} - C_{gd}$





$$\omega_{z\,eq} = \frac{1}{C_{eq} R_{Gz}} \qquad \omega_{z\,gs} = 0$$

C_{eq} cortado y C_{gs} abierto

$R_{eq} = R_{Gz}$ $\qquad R_{gs} = R_S$ ✗

C_{eq} abierto y C_{gs} cortado ✓

$R_{eq} = R_{Gz} \,//\, R_S$ $\qquad R_{gs} = R_{Gz} + R_S$

$\omega_{p\,eq} = 6678{,}3305 \frac{rad}{s}$ $\qquad \omega_{p\,gs} = \frac{1}{C_{gs} R_{gs}}$

$$C_{eq} = \frac{1}{\omega_{p\,eq} \cdot R_{eq}}$$

$\omega_{z\,eq} =$